

ArkLib’s KZG evaluation-binding theorem is axiom-clean **and** carries no information at any parameter: its t-SDH assumption quantifies over an unrestricted adversary type, which a `Classical.choice` trapdoor extractor inhabits with success probability exactly 1. We mechanize the refutation, the extraction-shaped repair that provably survives the exact attack, and the generic-group security bound — now mechanized **end to end** on ArkLib’s real `tSdhExperiment`, over the generic-restricted class, with the side-conditions named.

Research note — internal, not filed, not a security advisory: a **formalization-soundness** issue in a public, in-development library, not a vulnerability in any deployed system. KZG, t-SDH, and the reduction are sound as normally stated — the issue is a Lean quantifier. The central generic-group argument is complete and mechanized end to end.

I · The finding: axiom-clean, and empty

The mechanism

KZG evaluation binding is proved in ArkLib by a correct, constructive reduction to the t-SDH assumption — whose adversary is a **plain**, **unrestricted function type**:

```
abbrev tSdhAdversary (D : ℕ) :=
  Vector G₁ (D+1) × Vector G₂ 2 → StateT unifSpec.QueryCache ProbComp (Option (ZMod p × G₁))

def tSdhAssumption (D : ℕ) (error : ℝ≥0) : Prop :=
  ∀ (adversary : tSdhAdversary D), tSdhExperiment D adversary ≤ error
```

`ProbComp` is a free monad over oracle queries: only `query` nodes cost anything; pure computation is free and no resource bound is imposed anywhere. The SRS includes the verifier `leg` (g_2, g_2^τ) , which determines τ whenever $g_2 \neq 1$, and ArkLib’s own `exists_zmod_power_of_generator` (`Algebra.lean`:105) makes the discrete log `Classical.choice`-definable. The adversary reads g_2^τ , recovers τ , and returns the t-SDH solution $(c = 0, g_1^{1/\tau})$ — zero oracle queries, success probability exactly 1.

```
noncomputable def tauExtractingAdversary (hg₂ : g₂ ≠ 1) (D : ℕ) : tSdhAdversary D :=
  fun srs => pure (some (0, g₁ ^ (1 / dlogOf hg₂ srs.2[1])).val))

theorem tSdhExperiment_tauExtractingAdversary : tSdhExperiment D (...) = 1
theorem not_tSdhAssumption (herr : error < 1) : ¬ tSdhAssumption D error
theorem tSdhAssumption_trivial_of_one_le (1 ≤ error) : tSdhAssumption D error
```

No content at any parameter

Below 1 the assumption is refuted; at ≥ 1 its conclusion is the triviality “a probability is ≤ 1 .” `binding` consumes the assumption in the last step of its `calc`, and its hypothesis `hpair : pairing g₁ g₂ ≠ 0` **forces** $g_2 \neq 1$ (a bilinear map kills the identity) — exactly what the extractor needs. So `binding`’s hypotheses are jointly unsatisfiable for any error below 1 (`binding_hypotheses_unsatisfiable`), and its conclusion is free at ≥ 1 . The sibling ARSDH assumption falls identically, taking `function_binding` with it. A canary — the giving-up adversary scores exactly 0 — confirms the experiment discriminates. The reduction itself is fully constructive and algebraically correct; the vacuity lives entirely in the assumption’s quantifier.

The pattern, not a typo

Reducing to a **different** base assumption does not escape. The natural q-DLOG assumption, stated in ArkLib’s own unrestricted-adversary idiom, is equally false below 1 (`not_qDlogAssumption`, same extraction, own canary). And ArkLib’s algebraic-group-model scaffolding confirms the disease from the other side: `AGM/Basic.lean` is a stub (its adversary’s `run` is literally `sorry`; zero theorems; orphaned) and is **unsound as written** — the adversary is a `ReaderT` over the concrete group table, so its outputs can still depend on discrete logs. Its author flags the open problem verbatim: “TODO: need to be sure this definition is correct” “How to make the adversary truly independent of the group description? It could have had G hardwired.” Any concrete-group assumption of the shape $\forall$ unrestricted adversary, $\Pr[\text{win}] \leq \varepsilon < 1$ is `Classical.choice`-false in this idiom.

The methodological point: axiom checks are blind to vacuity

`binding`, the refutation, and the winning adversary all print the **same** clean axiom closure:

```
#print axioms not_tSdhAssumption -- [propext, Classical.choice, Quot.sound]
```

No `sorryAx`, nothing any `#print axioms / #assert_axioms` gate would flag. **Axiom-clean and vacuous coexist**. The blindness is structural: an axiom check reports a proof term’s closure; it never asks whether the theorem’s **hypotheses** are jointly satisfiable, and any `def FooHard : Prop` used as a hypothesis is an assumption no axiom check inspects. Query-based bounds (`IsQueryBoundP`) do not help either — the winning adversary makes zero queries. The only reliable test is adversarial: try to **inhabit the assumption’s negation** — prove the floor false at its deployed parameters. We found the identical pattern in our own hardness floors first, in several places, before ever looking at ArkLib; we present this as a field lesson, not a dunk.

II · The repair, the number, and the end-to-end theorem

The mergeable de-vacuation (mechanized)

ArkLib’s reduction is already constructive, and the assumption is consumed at exactly one `calc` step — so split there. The unconditional prefix becomes the primary theorem; the original `binding` becomes a one-line corollary.

```
theorem binding_reduces_to_tSdh ... (adversary : KzgBindingAdversary ...) :
  bindingExperiment ... adversary
    ≤ tSdhExperiment g₁ g₂ n (bindingReduction ... adversary)
```

No assumption `Prop`: both sides are concrete probabilities, true at every parameter — nothing for `Classical.choice` to inhabit (+41/−14 in one file; whole tree builds, 2994 jobs; axiom-clean). It **provably survives the exact attack** (`repair_survives_attack`): in one `sorry`-free closure, the trapdoor adversary still refutes `tSdhAssumption` below 1 **and** the repaired bound holds regardless — the unsatisfiable premise removed, every step of the reduction kept, the one obligation a sound assumption class must discharge isolated.

The number: the generic bilinear group bound

Group elements become opaque handles carrying **ordinary** polynomials in $\mathbb{Z}_p[X]$ — **not Laurent**: group inversion negates the exponent, it never introduces X^{-1} . A winning $1/(X + c)$ is therefore unrepresentable, and a “win” forces the nonzero, degree- $\leq D + 1$ polynomial $F_\ell \cdot (X + c) - 1$ to vanish at the random τ . Simulation (identical-until-bad) plus Schwartz–Zippel yield Boneh–Boyer’s Theorem 12, verified line by line against the source:

$$\varepsilon \leq (q_G + D + 3)^2 \frac{D + 1}{p - 1} = O\big((q_G + D)^2 \cdot D/p\big)$$

Not a clean q^2/p : the bound is cubic in the SRS degree D , and at production parameters the D^3/p term is the one to watch (Corollary 13’s $q < O(p^{1/3})$ side condition). Naive AGM is not a shortcut — a `Classical.choice` adversary returns a **valid** representation too; validity is not independence — it relocates the same content onto q-DLog’s generic hardness.

The end-to-end theorem (mechanized, sorry-free, no sorryAx)

The generic-group bound is now a **complete** mechanized theorem about ArkLib’s **own** `Groups.tSdhExperiment`, restated nowhere. For every generic strategy `strat` and query budget `fuel`:

```
tSdh_ggm_sound : tSdhExperiment D (embed strat) ≤ (C(fuel+D+4, 2)·D + (D+1)) / (p - 1)
```

the Shoup random-encoding number at $\delta = D$, a genuine < 1 in the standard regime (`tSdh_ggm_sound_lt_one`). **Why it escapes the vacuity**: it does **not** quantify over the full `tSdhAdversary` type — over which the statement is provably **false** (Panel I) — but over the **image of the generic embedding** `embed`: a strategy receives only equality booleans, never a group element, so it realizes only $g_1^{f(\tau)}$ with $\deg f \leq D$ — exactly what the counting bound bounds. `#print axioms` on the capstone and its full spine: `[propext, Classical.choice, Quot.sound]`. **Honest side-conditions, named** — “sound” means sound under these: $1 \leq D$ (genuinely **false** at $D = 0$: with no pairing, a G_1 adversary cannot form g_1^τ); $2 \leq p$; `orderOf g₁ = p` (with $g_1, g_2 \neq 1$); ArkLib’s own `SampleableType` instance; the < 1 regime $\binom{\text{fuel}+D+4}{2} \cdot D + (D + 1) < p - 1$.

The mechanized spine (all sorry-free, axioms [propext, Classical.choice, Quot.sound])

- **Vacuity refutations** — t-SDH, ARSDH, q-DLOG, each with a discriminating canary, against genuine ArkLib. **The repair and its survival** — `binding_reduces_to_tSdh`, `repair_survives_attack`.
- **The counting bounds** — static core `ggm_tSdh_sound`: $\varepsilon \leq (D + 1)/(p - 1)$ over the **entire** committed-generic type (the trapdoor extractor is untypable here; as far as our census found, the first generic-group security theorem in Lean); Maurer explicit-equality `adaptive_ggm_sound` (identical-until-bad **proven by induction**, not assumed); Shoup random-encoding `rand_encoding_bound` (all-pairs bad event and table size are theorems; its $\delta = D$ specialization feeds the capstone).
- **Degree discharge on the ACTUAL oracle** — ArkLib’s `tSdhAdversary` is granted **no pairing map**, so the oracle is purely linear; `hdeg_out_of_run / hdeg_handles_of_run` prove $\text{natDegree} \leq D$ by induction on the **real** `runAux/runTable` recursion — the degree hypotheses are theorems, the $\delta = D$ bound hypothesis-free.
- **The wiring** — `embed / embed_run_correspondence` (the group run steps in lockstep with the symbolic run); `experiment_eq_count` (ArkLib’s game collapses to `(winSet.card)/(p - 1)` — no monad plumbing left); `groupWinSet_eq_realWinSet` (the win predicate is ArkLib’s real `tSdhCondition`, by prime-order injectivity).

Off the critical path (optional — gates nothing)

The pairing-aware $\delta = 2D$ variant (`degree_invariant_paired`; the naive flat-table $2D$ claim **refuted**, X^4 at $D = 1$) is for a **pairing-endowed** oracle the ArkLib adversary does not have — the honest ceiling for a stronger, off-interface adversary. And re-typing `bindingReduction`’s adversary as a `Strat` — a convenience, not a gap: the result already holds for the whole `embed` image.